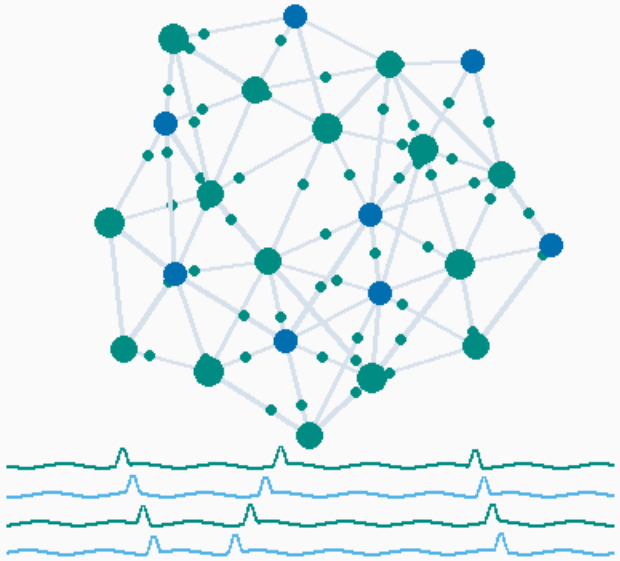
More information from fewer measurements

The challenge: measuring how a substance affects biological neural network development.

NEURONAL CULTURE + ELECTRODES



CONCENTRATIONS × DEVELOPMENTAL TIME POINTS



More concentrations and time points require more experimental work.

CONCEPTUAL DIAGRAM

Context: EPA Network Formation Assay

From three concentrations to a complete response

The context includes measurements at the time points covered by the study.

AVAILABLE MEASUREMENTS

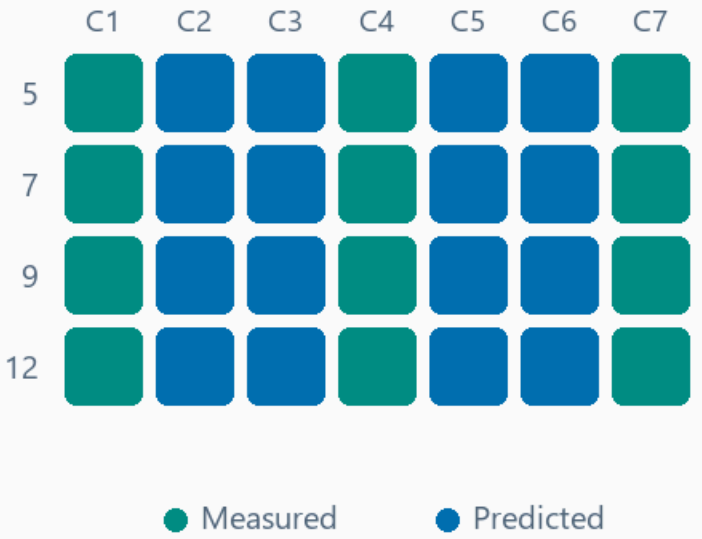




Digital twin

Learning across compounds

COMPLETED RESPONSE



Predicts responses at unmeasured concentrations and reports uncertainty.

Learning that informs experimental decisions

Learn across compounds, incorporate observations, and guide the next measurement.

01

Learning across compounds



Patterns shared across compounds.
A new compound needs
only a partial set of observations.

02

Integrated interpolation



Nearby measured doses
combined with learned patterns.
The model learns how to weight them.

03

Decision support

- Predict
- Assess confidence
- Choose a measurement

Curves and uncertainty intervals.
Activity, next-dose selection,
and investigation of early stopping.

Fair comparison: every method receives the same available measurements.

Building on earlier research, validating with real data

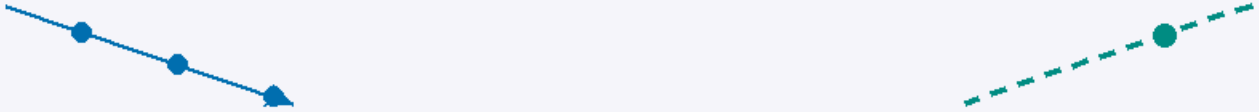
The project's origins and the conceptual link to early filtering.

NeuroChip Twin · initial prototype

Temporal phenotyping with synthetic data.
The starting point for the current project.

Speaking to Silicon

Using partial signals to filter costly work.
Inspiration for the early-stopping study.



NeuroChip Twin · submitted version

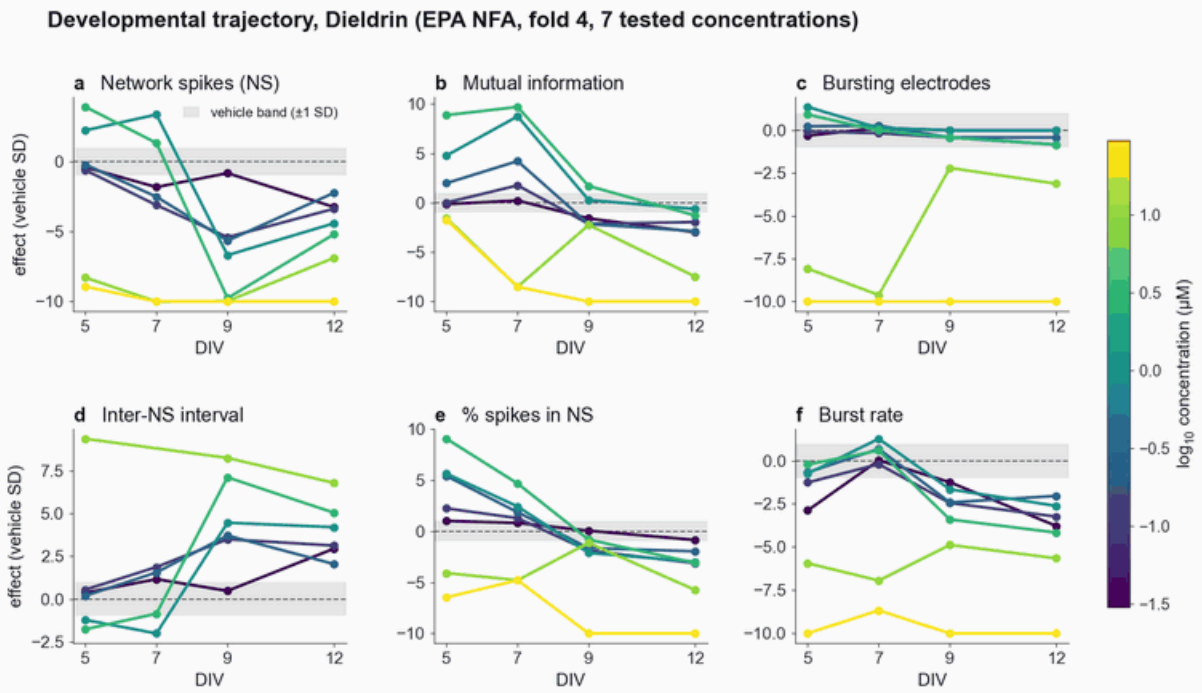
Core model: NeuroTrajectory · Real biological data · Evaluation by compound

The submitted version introduces new models and evaluation using real biological data.

Sources: NeuroChip v1-synthetic · Speaking to Silicon, arXiv:2601.12032

Real biological data

EPA Network Formation Assay: rat neuronal cultures with electrical readouts.



243
compounds

17
neural activity features

4
time points: days 5, 7, 9, 12

One compound is shown here; the main performance result is evaluated across compounds.

Evaluation on compounds unseen during training

Five compound-level folds: each test compound is excluded from its models' training data.

ROTATING THE OUTER TEST FOLD

	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
Round 1	Test	Train	Train	Train	Train
Round 2	Train	Test	Train	Train	Train
Round 3	Train	Train	Test	Train	Train
Round 4	Train	Train	Train	Test	Train
Round 5	Train	Train	Train	Train	Test

Additional robustness check

Exclude fold 0, which was inspected during development.

194

compounds in folds 1–4

−11.25%

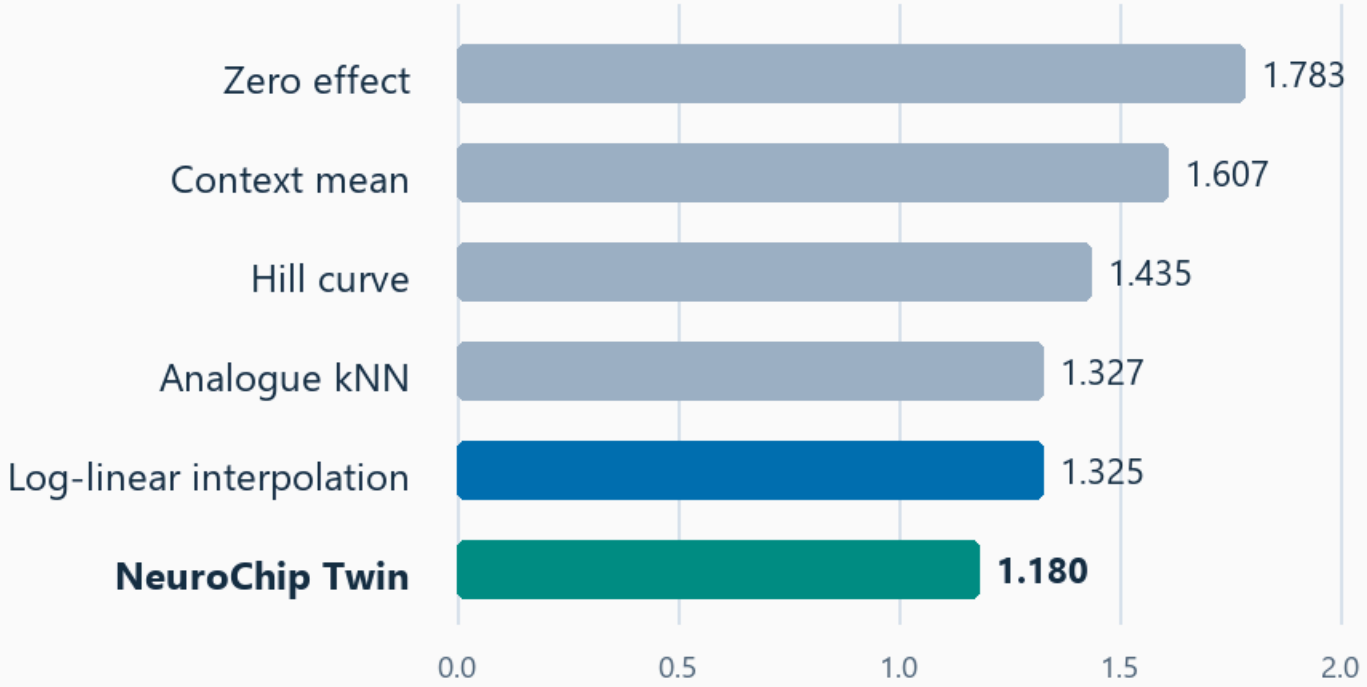
error relative to the baseline

All samples from each compound remain in the same fold.

Source: results/trajectory_cv.json · compound-level cross-validation

Lower error than five evaluated baselines

Measurements at three concentrations · Curve mean absolute error · Lower is better.



VERSUS THE STRONGEST BASELINE

−10.96%

mean absolute error
at unmeasured concentrations.

Paired difference

95% CI: −0.188 to −0.099

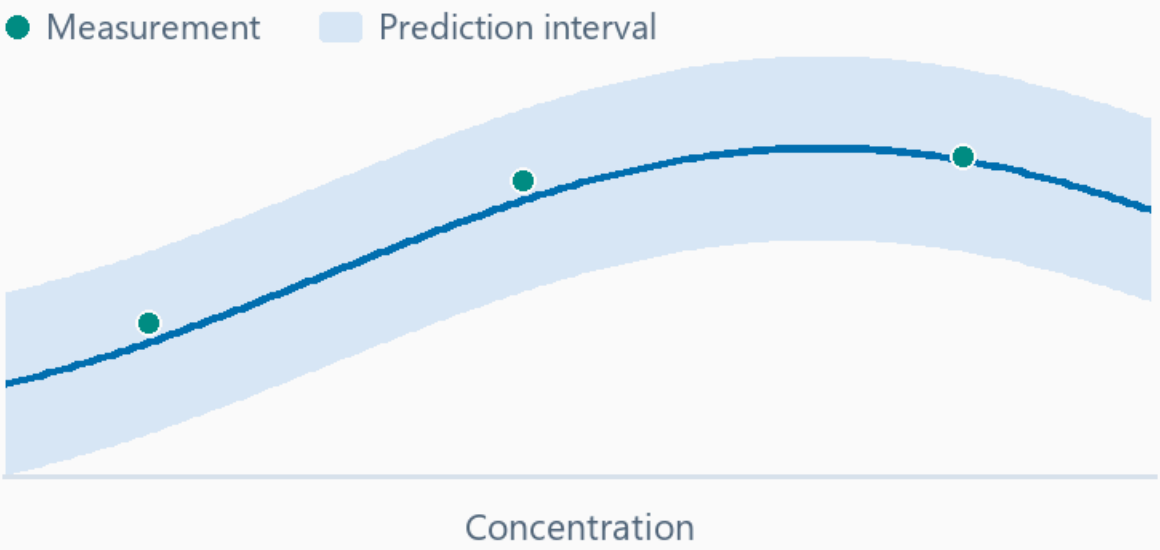
The interval lies entirely below zero.

Same input information for every method. MAE is in normalized assay units.

Predictions with quantified uncertainty

Conformal calibration: evaluating the coverage of prediction intervals.

READING A PREDICTION INTERVAL



Illustrative curve; not a prediction for a specific compound.

NOMINAL TARGET

90%

OBSERVED COVERAGE

89.9%

95% CI: 89.14–90.60%

Coverage evaluated retrospectively.
Calibrated across cytotoxicity regimes.

Predictive uncertainty guides which concentration to measure next.

An opportunity to reduce experimental work

A seven-concentration design with the same number of replicates at each concentration.



7 → 3

measured concentrations

−57.1%

treated wells

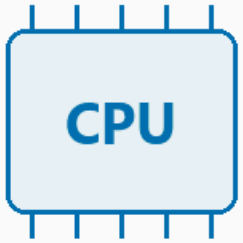
Same number of replicates at each measured concentration.

Measuring three of seven concentrations: 57.1% fewer treated wells.

From retrospective results to laboratory validation

An open, runnable system with a clearly defined next experimental step.

AVAILABLE TODAY

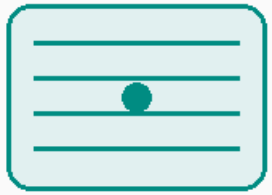


Models and demonstration.

CPU-based operation.

Public code and documentation.

NEXT STUDY



Select measurements with the system.

Run the laboratory experiments.

Compare predictions with results.

FUTURE EXTENSIONS



New chips and human cells.

Platform-specific validation.

Uncertainty recalibration.

Turning existing data into better-informed experimental decisions.

Validated results: what the data demonstrate

Retrospective cross-compound evaluation - real biological measurements

LOWER PREDICTION ERROR

10.96%

MAE 1.1797 vs 1.3249

243 unseen compounds; 3 doses

INTERVAL COVERAGE

89.9%

90% nominal prediction interval

Cross-conformal; 3 doses

BALANCED ACCURACY

86.4%

Hazard assessment on 97

shared test compounds

FEWER TREATED WELLS

57.1%

Potential design reduction

7 concentrations -> 3

EVIDENCE-LED INTERPRETATION

The error reduction is supported by a compound-paired 95% bootstrap CI [-0.1882, -0.0988].

77% of compounds benefited versus the strongest evaluated baseline at the three-dose budget.

The 57.1% figure is a design calculation, not a claim of measured cost savings in a laboratory.

POST-FINAL ANNEX - not a slide claimed to have been shown during the final presentation.

Better forecasting across evaluated baselines

Curve MAE at three measured concentrations; lower is better



-10.96% vs the strongest evaluated baseline

Paired MAE difference: -0.1452 | 95% bootstrap CI: [-0.1882, -0.0988] | n = 243 chemicals

Source: results/trajectory_cv.json - by_k.3.curve_mae; paired chemical bootstrap.

What is validated, and what comes next

Separating real-data results, assay-design potential and prospective validation

VALIDATED WITH REAL DATA

EPA NFA: 243 compounds and 6,902 wells.

17 network features across developmental days

5, 7, 9 and 12; held-out compounds in 5 folds.

15 trained model instances (5 folds x 3 seeds).

METHOD AND UNCERTAINTY

Probabilistic regression, not an RNN/LSTM.

Conditional neural-process encoder/decoder,

with dose-local attention and calibrated intervals.

Nested model selection includes Gaussian and Laplace.

SEPARATE REAL CHIP EXPERIMENT

Brewer dataset: 21 recordings, 440 axons.

Directed-layer feed-forward MAE improved 9.0%

vs tunnel-mean baseline; evaluation on nine

NoStim recordings (bootstrap, 10,000 resamples).

NEXT SCIENTIFIC MILESTONE

Prospective laboratory testing of dose selection

and forecasts. Measure actual time, well and

resource savings; validate on new chip platforms.

Retrospective benefits are not clinical validation.

Evidence: results/{trajectory_cv,conformal_r7,dnt_r4,chiplayer_r8}.json